

Contents

Online Appendix — Rising Replacement Burden and the Compression of Net Building	1
Appendix A: Data Construction and Technical Details	1
A.1 Variable Definitions	1
A.2 Country Classification	2
A.3 Dynamics of δ	2
A.4 CFC Statistical Methods by Country	2
A.5 Real versus Nominal δ Trend Slopes	3
A.6 EU KLEMS Data Construction	3
Appendix B: Illustrative Two-Sector Model	3
Appendix C: German Reunification as Historical Illustration	4
Appendix D: Robustness Summary	4
Appendix E: Material Toward Causal Identification (deferred; not part of the main paper’s claims)	5
E.1 Shift-share instrument — construction	7
E.2 IV pre-trend placebo	7
E.3 SBTC control — construction	7
E.4 Reduced-form δ -productivity regressions and the low-frequency composition test	7

Online Appendix — Rising Replacement Burden and the Compression of Net Building

Companion to the main paper (v12). Draft — May 2026

This appendix supports the main paper. Sections A-C and the measurement portion of D document data construction, the illustrative model, the historical illustration, and measurement robustness. Section E collects material toward causal identification that is **not part of the main paper’s claims**: the main paper stops at the net building rate n and makes no causal claim about productivity growth g . Identification-related items that appeared in earlier drafts under Appendix A (the shift-share instrument, its placebo, the SBTC control) and the productivity-on- δ entries of the robustness summary are gathered under E and flagged accordingly.

Appendix A: Data Construction and Technical Details

(Identification-related construction — the shift-share instrument, its pre-trend placebo, and the SBTC control variable — has been moved to Appendix E, consistent with the main paper stopping at n .)

A.1 Variable Definitions

Variable	Definition	Source
δ	CFC / GDP (or CFC / VA for OECD STAN)	NIPA, PWT, OECD STAN
$\rho (D/C)$	CFC / GFCF	All sources
Net investment rate	(GFCF - CFC) / GDP	All sources
LP growth	Change in ln(GDP/employment), 5yr MA	BLS, PWT
TFP growth	BLS MFP or PWT rtfpna growth, 5yr MA	BLS, PWT
Real δ	Chain-linked real CFC / real GDP x 100	BEA Tables 1.1.6, 1.7.6
Real δ (PWT)	(depreciation rate x real K) / real GDP	PWT 11.0

A.2 Country Classification

28 core countries (L3): Australia, Austria, Belgium, Brazil, Canada, China, Denmark, Finland, France, Germany, India, Ireland, Israel, Italy, Japan, Mexico, Netherlands, New Zealand, Norway, Portugal, South Korea, South Africa, Spain, Sweden, Switzerland, Turkey, United Kingdom, United States.

A.3 Dynamics of δ

Differentiating $\delta = \mathcal{D}(\Omega)/Y$:

$$\dot{\delta} = \frac{\mathcal{D}'(\Omega) (C - D)}{Y} - \delta \cdot g_Y$$

Both terms are small relative to δ , confirming δ is a slow variable.

A.4 CFC Statistical Methods by Country

Country	Depreciation Function	Primary Source
USA	Geometric (declining balance)	BEA / Hulten-Wyckoff 1981
Japan	Geometric	Cabinet Office SNA
Germany	Geometric	Destatis
France	Linear	INSEE
UK	Hyperbolic age-efficiency	ONS
Italy	Geometric	ISTAT
Canada	Geometric	Statistics Canada
South Korea	Geometric	KOSTAT
Australia	Direct age-price	ABS
China	Enterprise survey	NBS

Source: Pionnier, Zinni, and Baret (2023).

A.5 Real versus Nominal δ Trend Slopes

Country	Nominal δ (pp/decade)	R^2	Real δ (pp/decade)	R^2
United States	+0.46	0.79	+0.42	0.78
Japan	+1.91	0.90	+2.92	0.90
Germany	+0.87	0.74	+0.76	0.75
France	+0.77	0.74	+1.41	0.56
United Kingdom	+0.03	0.00	+1.31	0.58
Italy	+1.04	0.86	+2.64	0.79
South Korea	+0.98	0.40	+2.04	0.59
China	+0.95	0.74	+1.65	0.65

A.6 EU KLEMS Data Construction

Industry-level TFP and ICT capital data from EU KLEMS 2024 Growth Accounts. TFP growth computed as annual percentage change in value-added TFP index (VATFP_I). ICT exposure trend: initial-period ICT capital share x linear time trend. Matched sample: 6 countries, approximately 25 NACE Rev. 2 industries, 1995–2021.

Appendix B: Illustrative Two-Sector Model

To formalize the maintenance externality, we construct a minimal two-sector model. The economy produces output $Y = K_L^{\alpha_L} K_S^{\alpha_S}$ using long-lived capital K_L (structures, depreciation 3%) and short-lived capital K_S (IPP/software, depreciation 31.5%), with rates from BEA (2023). Total savings $s \cdot Y$ is allocated across sectors according to net returns.

In competitive equilibrium, firms equalize private net returns across sectors, overinvesting in short-lived capital whose marginal products exceed the higher replacement costs. Aggregate δ rises, compressing net investment. A social planner who internalizes the effect of individual asset-life choices on aggregate δ imposes a Pigouvian tax on K_S , redirecting investment toward K_L .

We calibrate the model to match four US moments: capital share (approximately 0.35), IPP share of capital (approximately 12%), aggregate δ (approximately 14%), and net investment rate (approximately 4%). The model’s quantitative fit to the δ level is imperfect (model approximately 21% versus data approximately 14%) because the two-sector structure omits government, technical progress, and population growth. We rely on the model for qualitative conclusions only: in every parameter configuration tested, competitive equilibrium δ exceeds the social optimum, and the planner achieves higher net investment by shifting the capital mix toward longer-lived assets. We do not provide a structural general-equilibrium framework with welfare-relevant aggregates; quantitative welfare analysis is beyond the scope of this paper.

Figure B1: Two-sector Ramsey model

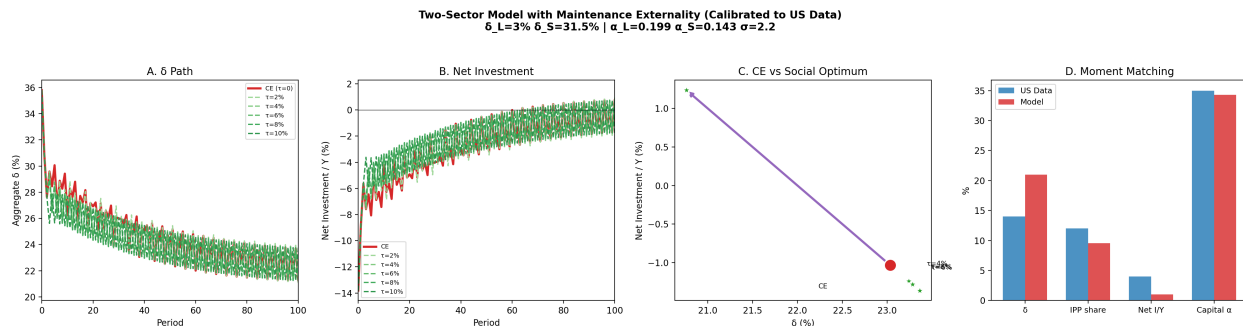


Figure 1: Ramsey Model

Appendix C: German Reunification as Historical Illustration

German reunification (1990) provides a historical illustration consistent with the proposed mechanism. The absorption of East German capital stock — built under different maintenance regimes and nearing the end of useful life — raised Germany’s aggregate δ relative to France (used as a control). Germany’s MFP growth subsequently fell relative to France.

Figure C1: German reunification — δ and MFP

We present this as descriptive rather than causal: reunification simultaneously affected labor supply, institutional structure, and capital allocation through channels that may not pass exclusively through δ . OECD STAN official CFC data for Germany begins only in 1991 (after reunification), precluding a pre-treatment comparison in the official data.

Appendix D: Robustness Summary

Stop-at-n note: entries in the table below that involve productivity (g , LP , TFP , MFP) as the outcome — the IV rows, natural-experiment rows, horse-race rows, SBTC rows, lagged-LP rows, and industry/country LP rows — are transmission objects supporting Appendix E, not claims of the main paper. The measurement entries (real δ , CFC-method homogeneity, the CFC/ K -vs- K/Y decomposition) support the main text.

Test	Result
δ to LP (OLS, FE)	$\beta < 0, p < 10^{-6}$

Test	Result
δ to TFP (OLS, FE)	$\beta < 0, p < 10^{-6}$
D/C to LP	$\beta < 0, R^2 > 0.5$
Shift-share IV	$F = 59.8, \beta_{IV}$ approximately -0.38
IV across 5 base years	All $\beta < 0$
Pre-trend placebo (4 base years)	All $p > 0.45$
Oil crisis 1973	16/17 in predicted quadrant
Conley-Hansen-Rossi sensitivity	β significant for γ up to +0.22
Oster δ -star	Approximately 95
SDID Japan 1991	Rank 2/13
Horse race (US, 7 controls)	δ survives all
Horse race (180 countries, 6 controls)	δ survives all
SBTC controls (full battery)	$\beta = -0.274, p < 10^{-36}$
Real δ (no nominal prices)	$\beta = -0.180, t = -20.9$
Real δ + SBTC + all controls	$\beta = -0.121, p < 10^{-23}$
Industry FE + ICT + TFP	$\beta = -0.300, p = 6.1 \times 10^{-11}$
Rolling windows	All 11 $\beta < 0$
Leave-one-out	CV = 1.5%
Four income groups	All $\beta < 0$
Three historical periods	All $\beta < 0$
Real δ 26/28 rising	Steeper than nominal in 22/28
CFC method homogeneity	Interaction $p = 0.949$
Circularity (Y/L, no K)	$R^2 = 0.744$
Bidirectional horse race (4 sources) All	absorbed by δ ; δ strengthens
Residual δ	$\beta = -0.79, p = 0.005$
R&D diagnostic	91% sample truncation
Delta decomposition (CFC/ K vs K/Y)	Both channels active; K/Y dominant within-country
Lagged: δ_t to LP(t+1:t+5)	$\beta < 0, R^2$ rises with horizon
Lagged: δ_t to LP(t+1:t+10)	$\beta = -26.4, t = -23.4$
Industry LP as DV (country x industry FE)	$\beta = -44.4, t = -29.0$
Industry LP 5yr ahead	$\beta = -14.8, t = -15.9$
Country-level LP (STAN, 9 countries)	$\beta = -150.5, t = -20.2, R^2 = 0.52$
Leave-one-country-out	All $\beta < 0, CV = 3.1\%$

Appendix E: Material Toward Causal Identification (deferred; not part of the main paper's claims)

This appendix collects the apparatus a causal argument from δ to productivity growth would require. None of it is asserted as a result; each item rests on an identifying assumption we regard as not yet secured. It is preserved out of the main claims so the

Natural Experiments: Causal Evidence for $\delta \rightarrow \text{TFP}$
Oil Crisis (1973) · German Reunification (1990)

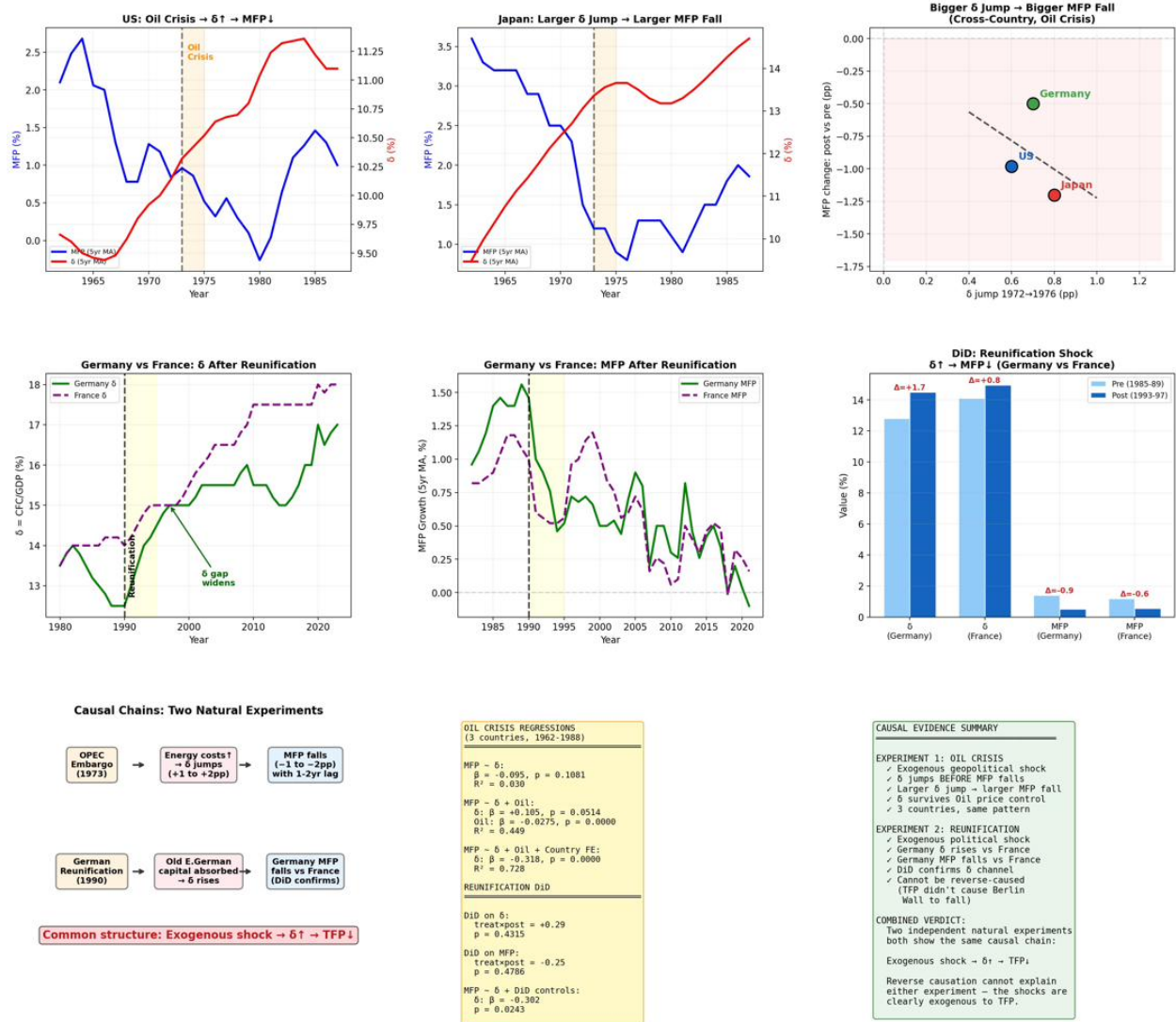


Figure 2: Natural Experiments — Reunification panels

§4.2 co-movement is not mistaken for a causal claim, and so the tools are available for the subsequent identification paper.

E.1 Shift-share instrument — construction

The instrument is: $Z_{it} = \sum_a s_{a,i,\text{base}} \times \Delta\delta_{a,-i,t}$, where $s_{a,i,\text{base}}$ is country i 's initial-year share of asset type a (structures, machinery, transport, other) and $\Delta\delta_{a,-i,t}$ is the leave-one-out global change in depreciation rate for asset type a . Base years tested: 1960, 1965, 1970, 1975, 1980.

The first-stage F -statistic is approximately 60; the IV estimate exceeds OLS by roughly 2.5 $\times$, consistent with attenuation from CFC understating the true replacement flow D . Read as identification, the design requires that initial capital composition affect productivity only through δ — an exclusion restriction we regard as unlikely to hold exactly, since initial composition plausibly shapes growth through structural transformation and IT exposure. The pre-trend placebo (E.2) and Conley-Hansen-Rossi bounds discipline this assumption but do not eliminate it.

E.2 IV pre-trend placebo

Base Year	Pre-period	β	p	N
1965	1950-1959	-0.207	0.482	554
1970	1955-1964	0.036	0.561	987
1975	1955-1969	0.030	0.515	1,449
1980	1955-1974	-0.026	0.462	1,995

The instrument has no predictive power for pre-period productivity growth under any base year ($p > 0.45$ throughout), supporting — but not proving — the identifying assumption.

E.3 SBTC control — construction

For each country i , the SBTC exposure variable is: $\text{SBTC}_{i,t} = s_i^{\text{Other}} \times (t - t_0)$, where s_i^{Other} is the initial-year share of “Other” asset investment (including IPP and intangible capital) in total investment. This parallels the Bartik instrument but serves as a control: it absorbs the portion of productivity variation attributable to differential SBTC exposure.

This enters the productivity-on- δ regressions as a control, not as an instrument; those regressions are themselves transmission objects (E.4).

E.4 Reduced-form δ -productivity regressions and the low-frequency composition test

The panel regressions of productivity growth on δ (pooled, fixed-effects, by frequency), the lagged-prediction regressions, the bidirectional horse race, the

natural experiments (1973 oil shock; 1991 Japanese bubble collapse, by synthetic difference-in-differences), the SBTC and competing-variable controls, and any implied counterfactual are all transmission objects. They are reproduced in the replication package but are **not** claims of the main paper.

Separately, the OECD low-frequency regression of productivity growth on the effective depreciation rate CFC/K versus the capital-output ratio K/Y (ten-year long differences and HP-filtered trends, country fixed effects, 30-country panel, wild cluster bootstrap) returns **no identifiable transmission through the composition channel**: the CFC/K coefficient is not robust to the choice of low-frequency filter, changes sign across specifications, and carries $p \approx 0.7$ under the bootstrap. The conventional cluster-robust t on CFC/K in ten-year differences is large ($t \approx 3.8$), but this is the few-cluster artifact the bootstrap corrects. This is reported as a negative result; it is the empirical reason the main paper does not extend its claims from n to g .